Final Project Report

Starting point:

We were provided with a dataset containing medical records of several patients, structured as described in the assignment PDF.
The objective is, in point 1, to predict different diseases (multiclass classification), and in point 2, to use a dictionary that allows us to determine whether a disease is chronic or acute and use the same dataset to perform a classification between these two types of disease (binary classification).

EDA:

The dataset contains 20,167 samples and 9 features.
By visualizing the data through plots, it is possible to observe several aspects: age is distributed in a more or less uniform way across the dataset, as is the number of symptoms. On the other hand, systolic blood pressure shows an irregular trend, as does stomach acidity.
However, these distributions do not seem very realistic for a real-world population. In fact, after checking external references, both age and systolic blood pressure usually follow a normal distribution, while the number of symptoms and stomach acidity tend to follow a right-skewed distribution.
Another unusual aspect is the number of individuals identified as *other* in the gender distribution, which appears surprisingly high in the dataset.
The target diseases, as well as the symptoms, do not show large differences in their distribution.
Moreover, the features are not strongly correlated with each other, as shown in the correlation matrix. Therefore, we can exclude the idea of removing features due to strong correlation.
From the box plots, we can see that all features are well distributed across quartiles, except for a few outliers, which were kept because their number is negligible compared to the size of the dataset.
The fact that the target classes are evenly distributed is very helpful, since in some cases models trained on highly imbalanced targets may tend to predict the class with the highest number of instances due to their internal design.
The dataset does not contain duplicate entries or missing values, so we can proceed with encoding.
During the encoding phase, after checking the distribution of non-numerical values and verifying that there were no anomalous categories, I decided to drop the first column containing useless indices, as well as the patient ID.
I converted the symptoms feature from a string into binary columns, and did the same for gender.
I then separated the target column after encoding it using a label encoder, since some models may have issues handling string targets.
As a result, I obtained:

- X: the cleaned dataframe containing all relevant features
- y: the target encoded using label encoding
- y_binary: the binary target encoded using the dictionary that will be used later

Multi-class classification:

Softmax Regression:

Metric / Parameter	Value	Notes
Optimal C Value	125	Determined via Grid Search; indicates moderate L2 regularization strength.
Cross-Validation Accuracy	42.65%	Mean accuracy across CV folds; provides a robust estimate of training stability.
Training Accuracy	50.82%	Accuracy on the complete training set (16,133 samples).
Test Accuracy	45.74%	Final accuracy on unseen data (4,034 samples); measures generalization.
Generalization Gap	5.08%	Difference between Train and Test accuracy, indicating a slight/moderate overfitting.

As the first model, Softmax Regression with ℓ2 regularization was implemented.
The hyperparameter C was tuned using Grid Search with 3-fold cross-validation. The optimal value found was C = 125. Since this is a relatively high value, the strength of the regularization is reduced, allowing the model weights to adapt more flexibly to the training data.
Although this approach leads to slight overfitting (a 5% gap between training and test performance), this level of flexibility proved necessary to handle the complexity of a dataset with 30 target classes. The model achieved a final accuracy of 45.74%. While this absolute value may appear low at first glance, it should be considered acceptable and statistically meaningful when compared to random guessing, which in a 30-class problem would yield a theoretical accuracy of approximately 3.3%.
If the value of C had been further increased beyond the optimal parameter (as tested in the code), the result would have been a model that does not generalize well, instead overfitting the training data and performing poorly on the test set.

Decision Tree Classifier:

Metric / Parameter	Value	Notes
Optimal Criterion	Entropy	Information gain sensitivity for better class separation.
Max Depth	10	Controlled depth to prevent over-specialization.
Min Samples Leaf	5	Ensures each rule is based on at least 5 patients.
Training Accuracy	83.09%	High fit on training data.
Test Accuracy	82.18%	High predictive power on new data.
Generalization Gap	0.91%	Minimal overfitting, very robust model.

We trained a Decision Tree model, and from the best-performing configuration we observed that entropy was the most effective splitting criterion. This is expected in cases where the target classes are uniformly distributed, as entropy often performs better than Gini in such scenarios.
The selected hyperparameters were max_depth = 10 and min_samples_leaf = 5, which were chosen by the code to maximize generalization. This resulted in a very small gap of only 0.91% between training and test performance, and a test accuracy of 82.18%.
The model demonstrated strong generalization capabilities and successfully avoided overfitting, thanks to the chosen hyperparameters.

Random forest:

Metric / Parameter	Value	Notes
Optimal n_estimators	200	Number of trees in the forest; higher values increase stability.
Optimal max_features	sqrt	Features considered at each split; ensures diversity among trees.
Optimal max_depth	None	Trees grew until leaves were pure or reached min_samples_split.
Min Samples Split	10	Minimum samples to split a node; provides ensemble-level regularization.
Out-of-Bag (OOB) Score	82.22%	Internal validation score; confirms consistency with test results.
Test Accuracy	82.85%	Final performance on unseen data (Generalization).
Overfitting Gap	6.38%	Moderate gap; indicates a high fit with good predictive power.
Leading Feature	Systolic BP	Most discriminative predictor with an importance of 30.06%.

We set n_estimators = 200, which makes the model more robust compared to a much smaller number of trees, which would have exposed the model to higher variance in the training data.
The number of features was set to sqrt, increasing diversity among the trees in the forest.
The min_samples_split parameter was set to 10 to ensure that each split in the tree is based on a statistically meaningful group of patients, avoiding the creation of overly specific rules for isolated cases.
The Out-of-Bag (OOB) score of 82.22% is very close to the test accuracy, indicating that the model generalizes very well and performs well even on unseen data.
Although the overfitting gap is relatively high—which is expected by nature in Random Forest models—the strong similarity between the OOB score and the test accuracy reassures us about the robustness and reliability of the model.

Gradient boosting:

Metric / Parameter	Value	Notes
Optimal Learning Rate	0.01	Controlled step size to avoid overshooting the minimum error.
Number of Trees	50	50 sequential trees optimized to correct previous errors.
Max Tree Depth	3	Individual trees are kept simple to ensure robustness.
Test Accuracy	82.87%	Excellent accuracy for a 30-class classification problem.
Overfitting Gap	0.11%	The lowest gap among all models tested.

Gradient Boosting builds models sequentially, where each tree attempts to learn from the errors of the previous one. This makes the training process relatively slow, since parallel computation is not possible.
In our optimal model, we used a learning_rate = 0.01, which is fairly low and indicates that the model learns gradually. Even with a relatively small number of trees (n_estimators = 50), the model achieved a good test accuracy of approximately 82.87%.
We set max_depth = 3, meaning that the trees composing the model are very simple. This does not imply poor performance, since the strength of Gradient Boosting does not lie in the complexity of individual trees, but in their sequential combination. The low value of max_depth allowed the model to generalize well and may explain the very low level of overfitting, with a gap of only 0.11%.

Staking:

Component	Summary
Base Model (Level 0)	Softmax, Decision Tree, Random Forest, Gradient Boosting
Meta-Modell (Level 1)	Logistic Regression (L2 reg, C=0.5)
Blender Input	120 Prediction Probabilities
Final accuracy	82.87%
Data Splitting	60% Base Train / 20% Blender Train / 20% Final Test

The implemented Stacking architecture allowed us to consolidate the knowledge acquired by the individual models. By adopting a 60–20–20 split, we ensured that the meta-model (Logistic Regression) learned to combine predictions using data that had not been seen by the base models, thus preserving the integrity of the validation process.

An analysis of the blender weights shows that the system primarily relies on non-linear models (ensembles), while Softmax Regression is used as a marginal corrective component. Although the final accuracy of 82.87% matches that of the best single model, the Stacking approach provides greater diagnostic reliability by reducing the overall variance of the system and delivering a more robust classification across the 30 target pathologies.

MLP:

Metric / Parameter	Value	Notes
Neural Architecture	50 x 50	Optimized depth and width of the hidden layers.
Initial Learning Rate	0.01	Step size used by the ADAM optimizer.
Selected Batch Size	64	Frequent weight updates for better convergence.
Test Accuracy	80.32%	High predictive accuracy on the 30 classes.
Generalization Gap	0.62%	Indicates no overfitting (Excellent status).
Total Parameters	~5,700	Number of trainable weights and biases.

A Multi-Layer Perceptron (MLP) was implemented using the ADAM optimizer and the ReLU activation function, as required by the assignment. Although the final accuracy of 80.32% is slightly lower than that of the forest-based models, this model demonstrated the best overall stability among all tested approaches.

Hyperparameter Tuning and Architecture

Architecture (50, 50):
The code selected a structure with two hidden layers of 50 neurons each. This depth was well suited to capturing non-linear relationships between symptoms and blood pressure without making the network excessively complex or difficult to train.

Learning Rate (0.01):
A learning rate of 0.01 was used. Thanks to the ADAM optimizer, this value allowed the network to converge rapidly toward the loss minimum without overshooting optimal solutions, stabilizing after only 21 iterations.

Batch Size (64):
Among the tested values (64, 128, 256), the model selected the smallest one. This proved beneficial for regularization: using batches of 64 introduces a moderate amount of stochastic noise during training, helping the network avoid overfitting the training data and improving generalization.

Overfitting and Convergence Analysis

The most significant result is the train–test gap of only 0.62%, which is almost negligible and indicates an extremely robust model.

The network automatically determined when to stop training (at the 21st epoch), preventing unnecessary training and reducing the risk of overfitting.
The loss curve shows a very smooth and stable descent, indicating that the chosen hyperparameters are well balanced and that the model learned in a consistent and controlled manner.

Binary classifiaction:

From the EDA for the binary classification, we observe that the two target classes are distributed as 36.6% Acute and 63.4% Chronic, indicating that the Chronic class is almost twice as frequent as Acute.

The symptoms appear to be reasonably well associated with both targets, with some symptoms being more strongly linked to one class than the other. However, no severe imbalance emerges that would strongly bias the classification process.

Logistic regression (binary):

Metric / Parameter	Value	Analysis
Optimal C	1	Found via 5-fold CV; provides ideal L2 penalty.
Binary Accuracy	63.19%	Accuracy for Acute vs Chronic classification.
Precision (Chronic)	0.66	Reliability of the "Chronic" prediction.
Recall (Chronic)	0.89	89% of chronic cases were correctly identified.
Recall (Acute)	0.19	Only 19% of acute cases were correctly identified.
Overfitting Status	None	Perfect alignment between Train and Test sets.

Binary Logistic Regression—which can be seen as the binary counterpart of Softmax Regression—achieved a stable 63.19% test accuracy with virtually no overfitting (0.14% train–test gap). However, the model is strongly biased toward the Chronic class, with a Recall of 89% compared to only 19% for Acute cases.

This imbalance is mainly due to the higher prevalence of Chronic samples in the dataset, combined with the intrinsic limitation of linear models in distinguishing acute conditions whose symptoms often overlap with chronic ones. As a result, while the model is highly reliable in identifying chronic patients, it shows poor sensitivity when detecting acute cases.

Random forest (binary):

Parameter / Metric	Value	Analysis
Optimal n_estimators	200	Large ensemble for high prediction stability.
OOB Score	94.56%	High internal validation consistency.
Test Accuracy	95.54%	Excellent ability to distinguish Acute vs Chronic.
Recall (Acute)	0.93	Significant improvement in detecting acute cases.
Recall (Chronic)	0.97	Almost perfect identification of chronic diseases.
Generalization Gap	4.46%	Well-regularized model despite 100% training score.

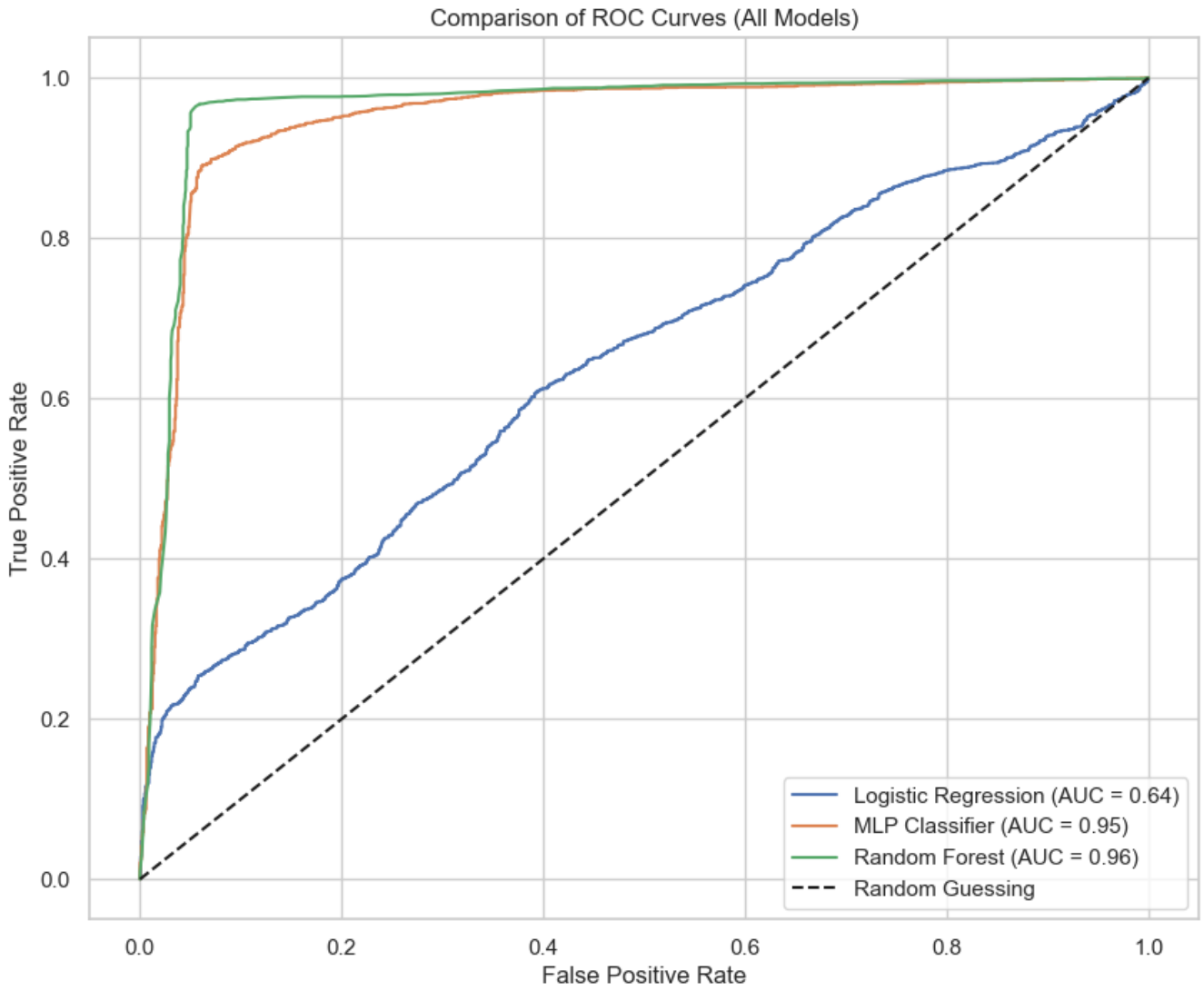
The Random Forest model drastically outperformed the Logistic Regression, achieving a 95.54% test accuracy. The key strength of this ensemble approach lies in its ability to handle the non-linear relationships between symptoms and disease nature. Unlike the previous linear model, the Random Forest successfully balanced the classification, reaching a 93% Recall for Acute cases, effectively overcoming the initial class imbalance. The strong correlation between the OOB Score (94.56%) and the test results confirms that the model is highly reliable and generalizes effectively to new clinical data.

MLP (binary):

Parameter / Metric	Value	Analysis
Optimal Architecture	(50, 50)	Balanced depth for feature extraction.
Best Batch Size	64	Efficient weight updates and noise-based regularization.
Test Accuracy	91.00%	High performance on unseen binary data.
Recall (Acute)	0.89	Strong ability to detect acute diseases.
Recall (Chronic)	0.92	Excellent reliability for chronic cases.
Generalization Gap	1.55%	Highly stable model with no significant overfitting.

The MLP Neural Network achieved a strong 91.00% test accuracy, significantly improving upon the linear Logistic Regression's performance. By utilizing two hidden layers of 50 neurons each and a ReLU activation function, the network successfully modeled the non-linear boundaries between Acute and Chronic classes. The Recall for Acute cases reached 89%, demonstrating that the model effectively mitigated the class imbalance issues. With an accuracy gap of only 1.55%, the MLP stands out as a highly generalized and reliable diagnostic tool, second only to the Random Forest in overall predictive power.

ROC curves:



The ROC curve comparison provides a definitive evaluation of our binary classifiers. The Random Forest and MLP Classifier exhibit exceptional diagnostic power, with AUC values of 0.96 and 0.95 respectively. Their steep ascent towards the upper-left corner demonstrates an excellent trade-off between sensitivity and specificity. In contrast, the Logistic Regression (AUC = 0.64) performs significantly worse, barely outperforming a random baseline. This visual evidence confirms that non-linear models are essential to distinguish between Acute and Chronic conditions in this medical dataset, as they can effectively navigate the complex overlap of symptoms.